

AMENDMENT NO. 1 JUNE 2026

TO

IS 5610 : 2025 REFRIGERANTS — SPECIFICATION

(Third Revision)

(Page 5, Table 2, Col. 3, 4, 6, and 8) — Substitute the following for the existing:

Sl No.	Characteristic	R-22 Technical	R-22 Pure	R-32	R-125
(1)	(2)	(3)	(4)	(6)	(8)
(i)	Moisture, ppm, w/w, <i>Max</i>	10	10	10	10
(ii)	Non-absorbable gases in vapor phase, percent v/v, <i>Max</i>	1.5	0.5	1.5	1.5
(iii)	*Initial boiling point, °C, <i>Max</i>	- 40.8	- 40.8	- 51.7	- 48.1
(iv)	*Boiling range from 5 to 85 percent evaporation, °C, <i>Max</i>	± 0.3	± 0.3	± 0.3	± 0.3
(v)	Acidity, as hydrochloric acid (HCl), ppm, w/w, <i>Max</i>	1	0.1	1	1
(vi)	High Boiling Residue, %, w/w, <i>Max</i>	0.002 0	0.001 0	0.010 0	0.010 0
(vii)	Appearance	Colorless	Clear and Colorless	Colorless	Colorless
(viii)	Particulates / Solids	Visually Clean	Visually Clean	Visually Clean	Visually Clean
(ix)	Chloride, no visible turbidity (indicates about 3 ppm)	No visible turbidity	No visible turbidity	No visible turbidity	No visible turbidity
(x)	Purity, %, w/w, <i>Min</i>	99.90 (B-11)	99.98 (B-11)	99.90 (B-11)	99.50 (B-10)
(xi)	All other volatile impurities, %, w/w, <i>Max</i>	0.10 (B-11)	0.02 (B-11)	0.10 (B-11)	0.50 (B-10)
(xii)	Halogenated unsaturated volatile impurities, %, w/w, <i>Max</i>	-	-	-	-
(xiii)	R-40, %, w/w, <i>Max</i>	-	-	-	-
(xiv)	CFC-12, %, w/w, <i>Max</i>	-	-	-	-
(xv)	HCFC-21, %, w/w, <i>Max</i>	-	-	-	-

(xvi)	HFC-23, %, w/w, <i>Max</i>	-	-	-	-
(xvii)	1-Chloro-1,1,2,2,2-Pentafluoroethane (R115), % w/w, <i>Max</i>	-	-	-	-
(xviii)	HCFC-31, %, w/w, <i>Max</i>	-	-	-	-
(xix)	HFC-32, %, w/w, <i>Max</i>	-	-	-	-
(xx)	HFC-134a, %, w/w, <i>Max</i>	-	-	-	-
(xxi)	Isomer HFC-134, %, w/w, <i>Max</i>	-	-	-	-
(xxii)	Total CFCs, %, w/w, <i>Max</i>	-	-	-	-
(xxiii)	Total HCFCs, %, w/w, <i>Max</i>	-	-	-	-
(xxiv)	Total Unsaturated Compounds, %, w/w, <i>Max</i>	-	-	-	-
(xxv)	Total Fluorinated Unsaturated Compounds, %, w/w, <i>Max</i>	-	-	-	-
(xxvi)	Sum HCC-30 (MDC) + HCC-40 (Methyl Chloride), %, w/w, <i>Max</i>	-	-	-	-
(xxvii)	HCFC-133a, %, w/w, <i>Max</i>	-	-	-	-
(xxviii)	1-Chloro-1,2,2,2-Tetrafluoroethane (R124), %, w/w, <i>Max</i>	-	-	-	-
(xxix)	Isomer ea R-245ca, R-245cb, R-245ea, R-245eb, %, w/w, <i>Max</i>	-	-	-	-
(xxx)	Odor	-	No Strange Odour	Slight Ether / No Malodor	-
(xxxi)	*Critical Temperature, °C	96.2	96.2	78.1	66

(Page 7, Table 3, Col. 3) — Substitute the following for the existing:

Sl No.	Characteristic	R-410A
(1)	(2)	(3)
(i)	Moisture, ppm, w/w, <i>Max</i>	10
(ii)	Non-absorbable gases in vapor phase, percent v/v, <i>Max</i>	1.5

(iii)	Acidity, as Hydrochloric Acid (HCl), ppm, w/w, <i>Max</i>	1
(iv)	High Boiling Residue, %, w/w, <i>Max</i>	0.001 5
(v)	Appearance	Colorless
(vi)	Particulates / Solids	Visually clean
(vii)	Chloride, no visible turbidity (indicates about 3 ppm)	No visible turbidity
(viii)	Purity, %, w/w, <i>Min</i>	99.9 (B-12)
(ix)	All other volatile impurities, %, w/w, <i>Max</i>	0.1 (B-12)
(x)	HFC-23, %, w/w, <i>Max</i>	-
(xi)	1-Chloro-1,1,2,2,2-Pentafluoroethane (R115), % w/w, <i>Max</i>	-
(xii)	HFC-32, %, w/w, <i>Max</i>	48.5 - 50.5 (B-12)
(xiii)	HFC-134a, %, w/w, <i>Max</i>	-
(xiv)	1-Chloro-1,1,2,2,2-Tetrafluoroethane (R124), %, w/w, <i>Max</i>	-
(xv)	1,1,1,2,2-Pentafluoroethane (R125), %, w/w, <i>Max</i>	49.5 - 51.5 (B-12)
(xvi)	1,1,1-Trifluoroethane (HFC-143a), % w/w, <i>Max</i>	-
(xvii)	Odor	Ethereal/sweet smell
(xviii)	*Bubble point, °C @ 101.3 kPa	-51.4
(xix)	*Dew Point, °C @ 101.3 kPa	-51.4
(xx)	*Critical Temperature, °C	71.4
(xxi)	R-600a, %, w/w	-
(xxii)	R-601a, %, w/w	-
(xxiii)	R-1234yf, %, w/w	-
(xxiv)	R-1234ze(E), %, w/w	-
(xxv)	R-116, %, w/w	-
(xxvi)	R-1336mzz(Z), %, w/w	-
(xxvii)	R-1130(E), %, w/w	-

(Page 15, Table 6, Sl No. **x**), and **xii**) — Substitute the following for the existing:

x)	Purity, %, w/w, <i>Min</i>	99.0 (B-10)	99.0 (B-10)	B-10
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xii)	Other Allowable Impurities, %, w/w, <i>Max</i>	0.5	0.5	B-10
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(Page 16, clause 6) — Substitute the following for the existing:

‘6 PACKING

6.1 The material (volume > 500 ml and up to 5 l) shall be supplied compressed and liquefied in cylinders with valve conforming to IS 7142: 2025 and the material having volume > 5 l shall be supplied compressed and liquefied in cylinders with valve conforming to IS 3196 (Part 2) of approved design and of suitable capacity and complying with the requirements of *Gas Cylinder Rules*, 2016 with such modifications as may be made from time to time by the Petroleum & Explosives Safety Organization, Government of India, or other duly constituted authority. In case the valve is purchased separately, it shall conform to either IS 8776, or IS 12300, or IS 3224, or IS 8737.

6.2 For the material having volume ≤ 500 ml, non-refillable metal container of approved design and of suitable capacity and complying with the requirements as mentioned in Table 7 shall be used. The non-refillable metal container shall pass the requirements of burst/pressure relief test as specified in 6.7.

Note — Cylinders/cans of 500 ml or less capacity are for one time use only and must be disposed after use.

6.3 For bulk supplies, ISO tanks may be used of approved design and of suitable capacity and complying with the requirements of *Static and Mobile Pressure Vehicles Rules*, 2016 with such modifications as may be made from time to time by the Petroleum & Explosives Safety Organization, Government of India, or other duly constituted authority.

6.4 The cylinders and ISO tanks shall be supplied with a tamper proof sealing and identification for each filling of the refrigerant gas.

6.5 The packing, marking, and labelling of cylinders shall be in accordance with the requirements of cylinders for liquid gases given in the *Gas Cylinder Rules*, 2016, with such modifications as may be ordered from time to time by the chief controller of explosives, Government of India, or any other duly constituted authority. Other containers/tanks shall also be marked and labelled in accordance with the instructions issued from time to time by the above-mentioned authority.

6.6 Refilling/ repacking from ISO tanks to cylinders or from big to small cylinders is allowed provided that the new packaging and the refrigerant conforms to the requirements given in this standard.

6.7 Burst/Pressure Relief Test

Burst/pressure relief test shall be performed on a sample size of 32 containers by subjecting them to hydrostatic test till rupture or leakage. The test shall be performed with the safety device installed on the containers. Burst pressure is the maximum pressure a vessel or component can withstand before it ruptures or leaks.

Each container shall be filled with water and suitably crimped at the mouth to enable a watertight seal during the test. The test apparatus shall contain a hydrostatic pump capable of developing a pressure at least 3 times the vapour pressure, and a pressure gauge of suitable range having at least count of 1.0 kg/cm².

The pressure shall be gradually increased till the container bursts or shows a leakage. The burst pressure shall be recorded for all 32 containers.

The C_{pk} value shall be calculated using the formula:

$$C_{pk} = (\mu - V_p)/3\sigma$$

where

μ = mean burst pressure for 32 observations;

σ = standard deviation of the 32 observations; and

V_p = designated vapour pressure of the gas at 65 °C.

The sample shall be considered meeting the burst/pressure relief test requirement if the C_{pk} value is more than 1.67, and the point of rupture was observed in the safety device or the bottom crimp, and not in the body in all the containers. Since the burst test is performed on empty containers, it can be done at supplier end and the test certificate may be provided with empty container consignment.’

(Page 17, Table 7, Col. (7)) — Substitute the following for the existing:

Sl No.	C_{pk} , Min
(1)	(7)
i)	1.67
ii)	1.67
iii)	1.67

(Page 20, sub-clause B-3.1.3.2) — Substitute the following for the existing:

‘B-3.1.3.2 Desiccant – any suitable desiccant.’

(Page 20, sub-clause B-3.1.3.12) — Substitute the following for the existing:

‘B-3.1.3.12 Syringe – 10 ml’

(Page 25, sub-clause B-4.3.4) — Substitute the following for the existing:

‘B-4.3.4 Calibration standard – any suitable certified calibration standard’

(Page 26, sub-clause B-5.1.2) — Substitute the following for the existing:

‘B-5.1.2 Thermometer – suitable thermometer with an accuracy of 0.1 percent and capable of measuring:

- a) – 80 °C to + 80 °C range for analysis of R32, and;
- b) – 50 °C to + 50 °C range for analysis of other refrigerants.’

(Page 32, clause B-11) — Insert the following note under clause B-11:

‘NOTES

1. The purity of R-22 may also be determined using the packed column gas chromatography (non-cryogenic) method. However, in case of any dispute, the capillary column gas chromatography (cryogenic) method shall be the referee method. The recommended GC operating conditions for packed column gas chromatography (non-cryogenic) method are given in Table 9.

2. The purity of R-32 may also be determined using the non-cryogenic capillary column gas chromatography method. However, in case of any dispute, the cryogenic capillary column gas chromatography method shall be the referee method. The recommended GC operating conditions for non-cryogenic method are given in Table 9.’

(Page 33, Table 9) — Substitute the following table for the existing:

Table 9 GC Operating Conditions

(Clauses B-11 and B-11.2)

2. Condition may need to be optimized for specific GC used.